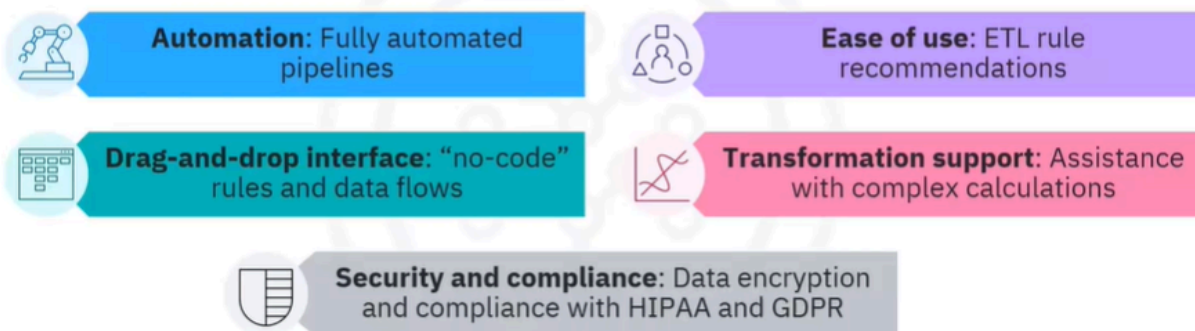


Data Pipeline Tools and Technologies

Features of modern pipeline tools

Typical enterprise grade technologies:



- There are many open-source and commercial data pipeline tools and Cloud services to consider. Typical features of modern enterprise-grade, ETL and ELT products include the following.
- Fully automated data pipeline creation from data extraction to loading in the destination, ease of use, rule recommendations for extracting, transforming and loading data some tools even crawl your data, a drag-and-drop GUI for specifying rules and data pipeline flows, also known as No-Code ETL, transformation support for and assistance with complex transformations, such as operations on strings calculations, and merging data, and security and compliance.
- Modern tools encrypt both data in transit and at rest and are certified compliant with industry and government regulations like HIPAA and GDPR.

Open source data pipeline tools

The Pandas Python library

- Versatile and popular programming tool
- Based on data frames – table-like structures
- Great for ETL, data exploration, and prototyping
- Doesn't readily scale to Big Data

Libraries with similar APIs: Vaex, Dask, and Spark help with scaling up

Consider SQL-like alternatives, such as PostgreSQL for Big Data applications

Skills Network

IBM

- Python, along with the Pandas library is a very popular and highly versatile programming environment for building data pipelines. Pandas uses a data structure called a data frame to handle Excel or CSV-style tabular data.
- It's a great tool for prototyping ETL pipelines and for exploratory data analysis but it can be challenging to scale to big data since data frame manipulations must be carried out in memory. Libraries with similar data frame APIs include Dask, Vaex, and Apache Spark, which can all help you to scale up to big data. For scalability, consider SQL-like alternatives to data frame APIs, such as PostgreSQL.

Open source data pipeline tools

Apache Airflow and Python

- Versatile “configuration” as code data pipeline platform
- Open-sourced by Airbnb
- Programmatically author, schedule, and monitor workflows
- Scales to Big Data
- Integrates with cloud platforms

- Apache Airflow, another package based on the Python programming language is a highly versatile and well-known example of an open-source configuration as code data pipeline platform. Apache Airflow was open-sourced by Airbnb and was created to programmatically author, schedule, and monitor data pipeline workflows. It was designed to be scalable and can handle an arbitrary number of parallel compute nodes, and Airflow integrates with most Cloud platforms, including AWS, IBM, Google Cloud, and Microsoft Azure.

Open source data pipeline tools

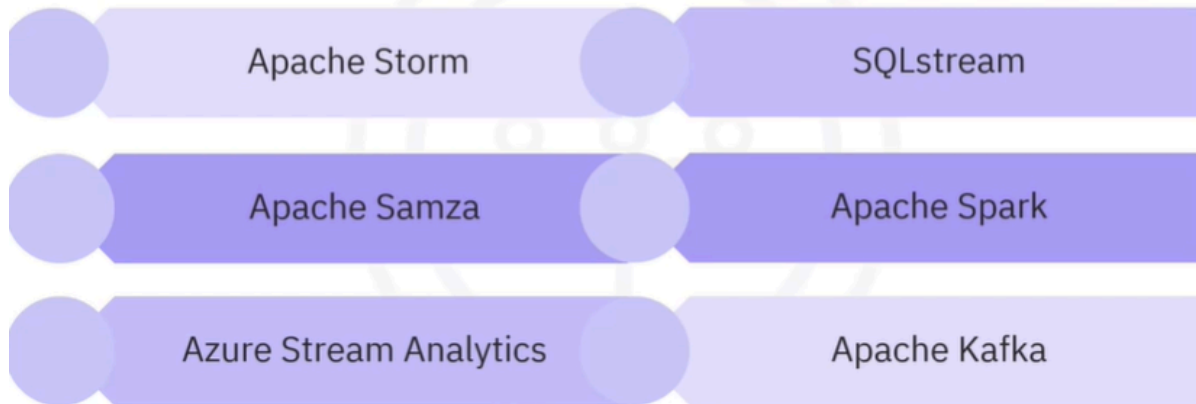
Talend Open Studio

- Supports big data, data warehousing and profiling
- Includes collaboration, monitoring, and scheduling
- Drag-and-drop GUI allows you to create ETL pipelines
- Automatically generates Java code
- Integrates with many data warehouses

- Talend Open Studio is yet another open-source data pipeline development and deployment platform. Talend Open Studio supports big data migration, data warehousing, and profiling, and it includes collaboration, monitoring and scheduling capabilities. It also has an interactive drag-and-drop GUI, which allows you to create ETL pipelines. There is no need to write code as Java code is automatically generated. It also connects to many data warehouses such as Google Sheets, RDBMS, IBM, DB2, and Oracle.

More streaming data pipeline tools

Stream-processing technologies include:



Recap

In this video, you learned that:

- Modern data pipeline technologies include schema and transformation support, drag-and-drop GUIs, and security features
- Pandas is an open-source Python libraries for prototyping and building data pipelines
- Apache Airflow and Talend Open Studio allow you to programmatically author, schedule, and monitor Big Data workflows
- Panoply is specific to ELT pipelines
- Alteryx and IBM InfoSphere DataStage can handle both ETL and ELT
- Stream-processing technologies include Apache Kafka, IBM Streams, SQLstream, and Apache Spark